

1. Administrative & Study Identification

Official Study Title: A Randomized, Double-blind, Double-dummy, Parallel-group Study, Comparing the Efficacy and Safety of Remibrutinib Versus Teriflunomide in Participants with Relapsing Multiple Sclerosis, Followed by Extended Treatment with Open-Label Remibrutinib **Brief Study Title:** Efficacy and Safety of Remibrutinib Compared to Teriflunomide in Participants With Relapsing Multiple Sclerosis (RMS) **Short Title/Acronym:** REMODEL-2 **Protocol Number:** CLOU064C12302 **NCT Number:** NCT05156281 **Posting ID:** 931878772591621536 **Sponsor Name / Company:** Novartis **Investigational Product:** Remibrutinib (LOU064) **Active Comparator:** Teriflunomide (Trade Name: Aubagio) **Phase of Development:** PHASE3 **Trial Status:** RECRUITING **Medical Monitor:** Novartis Clinical Operations / Medical Director

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To compare the efficacy and safety of remibrutinib versus teriflunomide in reducing the annualized relapse rate (ARR) in participants with relapsing multiple sclerosis (RMS). **Secondary Objectives:** To evaluate the time to 3-month and 6-month confirmed disability progression (3mCDP and 6mCDP) on the Expanded Disability Status Scale (EDSS), MRI lesion outcomes (e.g., new/enlarging T2 lesions), serum neurofilament light chain (sNfL) concentrations, and overall safety and tolerability.

2.2 Background & Rationale Remibrutinib is a novel, potent, and highly selective oral Bruton's tyrosine kinase (BTK) inhibitor. BTK is an enzyme crucial for the functioning of B-lymphocytes, which play a central role in the development of MS. This study aims to address the medical necessity for more effective and well-tolerated oral therapies for Relapsing Multiple Sclerosis by evaluating whether remibrutinib can provide superior prevention of MS relapses and delay disability progression compared to teriflunomide (an approved active comparator).

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator (Teriflunomide) **Blinding:** Double-blind, Double-dummy **Randomization:** Parallel-group

3.2 Planned Enrollment & Duration **Targeted Enrollment:** 800 participants **Number of Sites:** Global multi-center trial (approx.

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. 18 to 55 years of age. 2. Diagnosis of RMS according to the 2017 McDonald diagnostic criteria. 3. At least 1 documented relapse within the previous year, OR 2 documented relapses within the previous 2 years, OR 1 active Gadolinium (Gd)-enhancing lesion in the 12 months prior to screening. 4. EDSS score of 0 to 5.5 (inclusive). 5. Neurologically stable within 1 month.

4.2 Exclusion Criteria 1. Diagnosis of primary progressive multiple sclerosis (PPMS). 2. Disease duration of more than 10 years in participants with EDSS score of 2 or less at screening. 3. History of clinically significant CNS disease other than MS, or history of confirmed Progressive Multifocal Leukoencephalopathy (PML). 4. Ongoing substance abuse, history of malignancy, or suicidal ideation/behavior. 5. Active clinically significant systemic bacterial, viral, parasitic, or fungal infections (including positive results for syphilis, tuberculosis, or HIV). 6. Uncontrolled disease states, such as asthma or inflammatory bowel disease. 7. Active, chronic disease of the immune system other than MS (e.g., rheumatoid arthritis, SLE). 8. History or current treatment for hepatic disease or severe renal disease. 9. Abnormal hematology parameters at screening or significant ECG abnormalities (Resting QTcF ≥ 450 msec for males or ≥ 460 msec for females). 10. Pregnant or nursing (lactating) female participants. 11. Receipt of live or live-attenuated vaccines within 6 weeks of randomization.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Oral administration of remibrutinib tablet and a matching placebo capsule for teriflunomide (double-dummy design). **Route of Administration:** Oral **Comparator Details:** Teriflunomide (Trade Name: Aubagio, ATC Code: L04AA31) | EMA Info: [Aubagio EPAR](#) **Duration of Treatment:** Maximum duration per participant of up to 30 months in the Core Part, followed by an Extension Part of up to 5 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive MS therapies and symptom management not affecting immune function. **Prohibited:** Concurrent disease-modifying therapies for MS, strong immunosuppressants, systemic immune therapies (e.g., Methotrexate, Leflunomide), strong CYP3A4 inhibitors, or moderate/strong CYP3A4 inducers within two weeks prior to randomization.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -30 to 0	Informed Consent, Medical History, Baseline MRI, EDSS scoring, Labs
Baseline	Day 0	Randomization, First Dose of Core Treatment, Vitals, ECG
Interim	Every 3 months	Onsite Visits: Safety Labs (blood draws), Vitals, Efficacy Assessment (EDSS, Relapse Evaluation), Participant Diary Review
End of Core Study	Up to Month 30	Final Core Visit, IP Accountability, Transition to open-label Extension Part or Safety Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized relapse rate (ARR) of confirmed relapses during the Core Part. ARR is the average number of confirmed MS relapses in a year. **Measurement Method:** Clinical evaluation of relapses by a blinded independent rater from Baseline up to 30 months.

7.2 Secondary & Exploratory Endpoints * Time to 3-month confirmed disability progression (3mCDP) on the Expanded Disability Status Scale (EDSS) [Core Part] (pooled data). * Time to 6-month confirmed disability progression (6mCDP) on EDSS [Core Part] (pooled data). * Annualized rate of new or enlarging T2 lesions on MRI [Core Part]. * Serum neurofilament light chain (sNfL) concentrations.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Method of collection includes patient diaries, frequent lab draws (every 3 months), and physician assessments during onsite clinic visits. **Serious Adverse Events (SAEs):** Reported within 24 hours. Specific monitoring is in place for liver function and potential systemic infections. **Data Monitoring Committee (DMC):** An independent external Data Monitoring Committee evaluates unblinded safety data periodically to ensure patient safety across the global sites.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analysis; Safety Set for all AE/SAE evaluations. Data will be pooled with an identically designed sister study (CLOU064C12301 / REMODEL-1) for certain disability endpoints (e.g., 3mCDP and 6mCDP). **Sample Size Justification:** Targeted enrollment of 800 participants appropriately powered to detect a statistically significant reduction in ARR for remibrutinib compared to the

active comparator, teriflunomide. **Handling of Missing Data:** Advanced statistical modeling (e.g., negative binomial regression for ARR) and Multiple Imputation for missing continuous variables or early dropouts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring of investigator sites to ensure strict protocol adherence and double-dummy blinding integrity. **Audit Trail:** Centralized Electronic Data Capture (EDC) system with comprehensive audit trails, including independent central reading for MRI scans and independent blinded EDSS raters.

11. Study Identifiers & Governance

Primary ID: CLOU064C12302 **Registry Number:** NCT05156281 **Posting ID:** 931878772591621536 **Sponsor/Lead Organization:** Novartis **Study Title:** REMODEL-2 **Official Regulatory Title:** A Randomized, Double-blind, Double-dummy, Parallel-group Study, Comparing the Efficacy and Safety of Remibrutinib Versus Teriflunomide in Participants with Relapsing Multiple Sclerosis, Followed by Extended Treatment with Open-Label Remibrutinib

12. Clinical Framework (Multiple Sclerosis Specific)

Disease Area: Relapsing Multiple Sclerosis **Preferred UMLS Name:** Multiple Sclerosis, Relapsing-Remitting **MeSH Code:** C08.381.560.675 **SNOMED ID:** 24700007 **Diagnosis Threshold:** 2017 McDonald diagnostic criteria; EDSS score 0 to 5.5 **Medical Methodology:** Bruton's Tyrosine Kinase (BTK) inhibitor vs. active comparator (pyrimidine synthesis inhibitor) **Optimization Target:** Reduction in Annualized Relapse Rate (ARR) and confirmed disability progression

13. Study Procedures & Methodology

Management Pathway: Randomized, double-blind, double-dummy Core treatment followed by an open-label single-arm Extension **Education Protocol (HCP):** Standardized Investigator Training and EDSS Rater Certification **Education Protocol (Patient):** Double-dummy pill administration education and participant diary maintenance **Quality Audit Mechanism:** Centralized independent MRI reading and independent, blinded EDSS clinical evaluation

14. Observation & Follow-Up Schedule

Total Duration: Up to 30 months (Core Part) + up to 5 years (Extension Part) **Onsite Visit Cadence:** Every 3 months (Core Part) for vitals, blood draws, and physical/neurological exams **Remote Monitoring Frequency:** Between clinic visits as necessary for diary compliance and AE checks **Checklist Review Interval:** EDSS scoring and thorough neurological examinations at defined protocol milestones (e.g., every 3 to 6 months)

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				